CODE & CONSULT HACKATHON

Grind & Crumb Hackathon Client Brief

The Client & The Challenge

Grind & Crumb's board has £1.5m to invest next year and wants your team to tell them where it should go, backed by their own data.

Grind & Crumb is a fictional independent café chain with 12 stores across Sheffield, Leeds, York, Manchester, Nottingham, Harrogate and Hull. It sold about £8.7m over July 2025 to June 2026 across three channels: in-store, app click & collect, and two delivery platforms. The newest store, Leeds Trinity, opened in September 2025.

The CEO, Anna Whitfield, puts it like this:

> *"Sales are up, but profit doesn't feel like it. My managers each tell me a different story. Do we open more stores, fix the ones we have, push harder on delivery, or invest in the app? I need to see what's actually going on, store by store, and I need a tool my ops team can keep using after you leave".*

The board's options are not limited to that list. Strong teams will find problems the CEO hasn't spotted yet.

Deliverables

Each team picks a problem in the data, builds a working tool that solves it, and recommends how to spend the £1.5m.

1. **The tool.** Find a problem in the data and build a working web app or tool that solves it. A dashboard that gives the CEO visibility is one valid answer. So is a staffing planner, a delivery pricing calculator, a site selection model, a review theme tracker or a lapsed customer win-back tool. You choose the problem; judges score how well you've solved it.
2. **The recommendation.** Where the £1.5m goes, the expected payback, and what you would stop doing.
3. **The report.** A 2 page executive summary covering your findings, the £1.5m allocation, the numbers behind it and key assumptions, plus up to 2 pages of appendix. Submitted at the end of the build, before pitches start.
4. **The pitch.** 5 minutes plus 3 minutes of Q&A, delivered to the CEO and board (the judges).

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Marking Criteria

Criterion	Weight	What judges look for
Insight	30%	Root causes found and quantified, from numbers and qualitative sources, not just symptoms described
Recommendation	25%	Clear £1.5m allocation, numbers behind it, risks acknowledged
Tool	25%	Solves a real problem the data shows, usable by the client, works live
Report and pitch	10%	Clear, concise, answers the CEO's question in the first minute
Data rigour	10%	Data issues spotted and handled, comparisons fair, assumptions stated

(Use AI tools as much as you like, but be ready to explain every number)

Proposed Timeline

Time	What teams should be doing
0:00 - 0:30	Brief, data walkthrough, questions for the client
0:30 - 2:00	Explore the starter file and qualitative sources, pick your problem
2:00 - 5:30	Build the tool, dig into the raw files, test your recommendation
5:30 - 6:30	Write the report, prepare the pitch
6:30 - 7:00	Submit the report, final testing
7:00 - 8:00	Pitches and judging

Data Pack

The pack has three layers, so every team can start analysing in minute one, whether they work in Excel or code. **Everything covers 1 July 2025 to 30 June 2026.** Full column definitions are in the README inside the zip.

1. **Starter file.** `starter_store_monthly_pnl.csv` is a clean profit and loss table, one row per store per month. It opens straight in Excel and shows where the problems are.
2. **Raw data.** Twenty CSVs for finding out why. They join on `store_id`, `product_id`, `receipt_id`, `customer_id`, `date` or `city`.
3. **Qualitative sources.** Stakeholder interviews, platform emails, news notes, a staff exit survey, a customer survey and 3,030 customer reviews.

File	Rows	What it holds
starter_store_monthly_pnl.csv	142	**Clean** store x month P&L: sales by channel, every cost line, store profit, waste, staff turnover
stores.csv	12	Format, location type, size, seats, rent, opening date, manager
products.csv	38	Menu and delivery prices, unit cost, seasonal availability
transactions.csv	143,093	One row per receipt: store, time, channel, customer, totals **(10% sample)**
transaction_lines.csv	238,624	Items on each receipt, milk choice, modifiers, discounts
daily_store_sales.csv	4,292	**Full** daily receipts and net sales by store and channel
footfall.csv	12,876	Door counter by store, day and daypart
weather.csv	2,555	Daily temperature and rainfall by city
loyalty_customers.csv	9,000	Loyalty members: signup, home store, age band, opt-in

staff_rota_weekly.csv	1,875	Scheduled hours and wage by store, week and daypart
staff_monthly.csv	142	Headcount, leavers, starters, staff survey score
waste_log.csv	12,876	Food prepared and wasted by store, day and category
store_costs_monthly.csv	142	Rent, rates, utilities, repairs, other opex
capex.csv	4	Past fit-out and refit spend
promotions.csv	6	Promotions run, mechanics, stores, marketing spend
delivery_partner_terms.csv	3	Commission and packaging cost by platform
customer_reviews.csv	3,030	Free-text reviews with rating and source
store_catchment.csv	12	Footfall, offices, students, income, competitors per store
candidate_sites.csv	7	Possible new sites with the same catchment fields, rent and fit-out quote
business_events.csv	5	Key dated events: app releases, price changes, openings
qualitative/stakeholder_interviews.md	7 interviews	CEO, four store managers, a barista, Head of Digital
qualitative/delivery_platform_emails.md	3 emails	Dashly and RushEats correspondence
qualitative/news_and_market_notes.md	8 notes	Local news, competitor moves, market context

qualitative/staff_exit_survey.csv	115	Why staff left, by store, with comments
qualitative/customer_survey.csv	650	May 2026 loyalty survey with open text answers

Sample vs full. *The receipt-level files are a random 10% sample of all receipts. Use them for basket, product, time-of-day and customer questions. Use* `daily_store_sales.csv` *for total sales*

Assumptions & Ground Rules

Use these assumptions so every team's numbers are comparable.

- Treat all prices and sales as net of VAT.
- Labour cost = scheduled hours × hourly wage × 1.13 (employer on-costs).
- Delivery platforms take their commission on the order value; Grind & Crumb pays the packaging cost per order. Each delivery order also takes about 4 minutes of staff time to pack.
- Ignore head office costs and tax; compare stores on store-level profit.

What the managers say, for what it's worth:

- Leeds Briggate: *"The city centre is just quieter this year"*.
- Manchester Piccadilly: *"We're the biggest store in the chain, we're obviously the most profitable"*.
- Hull: *"We always run out, so I order extra to be safe"*.
- Head of Digital: *"The new app is a big upgrade, customers just need time to adjust"*.

The raw files contain a handful of real-world data issues; the starter file has already been cleaned. Spotting and handling the issues earns marks, but it shouldn't take more than half an hour. Ask the organisers if something looks impossible, but don't expect them to tell you what's wrong.